

# VAHAGN GHAZARYAN

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## EDUCATION

### Shanghai Jiao Tong University

Sep 2021 – July 2025

B.Sc. in Software Engineering

Relevant Courses: Assembly, Computer Organization, Design of Algorithms, Operating Systems, Advanced Data Structures, Compilers, Cloud Computing

## INDUSTRY EXPERIENCE

### Cognaize, LLC

Sep 2018 - Sep 2020

Data Engineer

- Developed AI-based ETL pipelines, integrated cloud storage systems, and built REST APIs for data access and management.

## RESEARCH EXPERIENCE

### Institute of Parallel and Distributed Systems, Shanghai Jiao Tong University

Undergraduate Research Assistant | Advisor: [Mingyu Wu](#)

Apr 2024 – Mar 2024

- Benchmarked Android applications co-running with LLM agents; contributed to literature review and profiling tool development.

## PROJECTS

- smart-madvise**  
A kernel module that dynamically provides memory management hints based on real-time application usage patterns.
- minik8s**  
Lightweight Kubernetes-like container orchestration system written in Go, featuring integrated CI/CD pipelines and comprehensive test coverage.
- Distributed File System**  
Inode-based distributed file system implemented in C, with support for concurrency, atomic operations, and crash recovery.
- Raft and MapReduce**  
Implementations of the Raft consensus algorithm and MapReduce model in C, based on the original research papers.
- Tiger Compiler**  
Full compiler for the Tiger programming language, covering all stages from lexical analysis to register allocation.
- Online Shop Platform**  
A scalable, fault-tolerant, low-latency online shopping platform built using React.js, Spring Boot, MySQL, MongoDB, Neo4j, Redis, Kafka, Apache Spark, Nginx, and Docker.
- eBPF-Based Security Framework**  
Runtime security framework using eBPF to dynamically monitor and control inter-process communication (IPC) channels like pipes and signals.

## AWARDS

- Third Prize, **China Postgraduate Innovation and Practice Series Competition (CPIPC)** Oct 2024
- First Prize, **Hack the DNA Datathon**, organized by ISTC & PMI Science Dec 2019

## SKILLS

**Core Competencies:** Linux kernel, Android Framework.

**Programming Languages and Tools:** C, C++, Python, Go, Java, Git, Tmux, LaTeX.

**Languages:** Armenian (Native), English (Fluent), Russian (Intermediate), Chinese (Intermediate).

**Professional Attributes:** Adaptability, Resilience, Teamwork, Public Speaking.